Proceedings of the 8th International Conference on Inventive Material Science and Applications (ICIMA-2026)
IEEE Xplore Part Number: CFP264B2-ART; ISBN: 979-8-3315-7814-5

# A Low-Latency IoT Framework for Real-Time Driver Fatigue Detection using YOLO and Multi-Level Safety Alerts

Dr. T. Senthil Kumar
Assistant professor
Electronics and communication Engineering
M.Kumarasamy College of Engineering
Karur, Tamilnadu.
senthilkumar.ece@mkce.ac.in

Indumathi R
UG Scholar
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, TamilNadu.
927622bec075@mkce.ac.in

Janani B C
UG Scholar
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, TamilNadu.
927622bec081@mkce.ac.in

Kavika V
UG Scholar
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, TamilNadu.
927622bec093@mkce.ac.in

Shobika G
UG Scholar
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, TamilNadu.
927622bec189@mkce.ac.in

2026 8th International Conference on Inventive Material Science and Applications (ICIMA) | 979-8-3315-7814-5/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICIMA68728.2026.11564223

***Abstract*—Driving fatigue is one of the major causes of road accidents especially in long distance journeys and in commercial services. They need to monitor the alertness of the drivers all the time in order to improve road safety and avoid the situation with driver fatigue. The paper intends to present a low-latency IoT-based scheme of fatigue detection in drivers in real-time through the use of deep learning and multilevel safety alerts. It records facial behavior with an onboard camera and classifies potential fatigue signs with an object detector model based on YOLOv11 and recognizes them as long eye closures and unusual head positioning. A microcontroller called ESP32 also coordinates the output of detection and triggers hierarchical alert devices, such as audible warning buzzers, vibration alerts, LCD display warnings, and external vehicle LEDs. To guarantee a minimum level of communication lag and quick reaction, the framework employs the User Datagram Protocol (UDP) in order to transfer data between the vision processing unit and the IoT hardware. The system has built in edge processing and buffering techniques to accommodate packet unnecessariness or network overcrowding. Also, remote monitoring is on with the help of automatic email notifications with taken pictures to the fleet manager or vehicle owner. Testing with a wide range of drivers of various ages, genders and in the real-world on-the-road situation has shown that the framework results in low response times, high detection rates and low cost of implementation and thus it is applicable in intelligent transportation systems and commercial fleet safety management.**

***Keywords—Driver Fatigue Detection, Internet of Things, YOLOv11 Algorithm, ESP32, Computer Vision, UDP Protocol, Road Safety.***

## I. INTRODUCTION

Drowsiness and distracted behavior that lead to road traffic accidents among drivers continue to be a significant issue of concern in terms of safety in the world, especially in the long-haul, and commercial vehicle use. The impairment that is caused by fatigue reduces the reaction time of the drivers, situational awareness and ability to make decisions, which cause serious and usually fatal crashes. Consequently, intelligent, real-time, non-intrusive driver monitoring system development has emerged as an important research area in intelligent transportation systems and advanced driver assistance systems (ADAS). The current innovations of computer vision and deep learning allow capturing driver behaviors on the camera in the cab. Deep learning models with the use of the YOLO technique have attracted much attention because of their high detection power and real-time operations. Poon et al. [1] have shown that multi-category driver behavior detection, one of them being drowsiness and head turning, is possible on embedded GPU platforms with real-time performance. Equally, Soe et al. [2] used YOLO to detect abnormal driving behaviour like phone use and drowsiness with a high level of accuracy and low computing delay. These works affirm the appropriateness of YOLO architectures in driver monitoring in real-time. More current literature has been aimed at enhancing detection accuracy and strength. Ma et al. [3] proposed an attention-assisted YOLOv8 model that concurrently detects distracted driving and driver emotion under better convergence and real-time results with Jetson Nano machines. A lightweight fatigue detection technique based on the system of YOLO that combines various facial features, including the state of the eye and mouth, was proposed by Zhao et al. [4], which showed a higher accuracy rate but with less struggle. Although these strategies have the benefit of improving the efficiency of models, they are mostly concerned with improving algorithms and have no system-level safety response systems. Another area of research that has been conducted pertains to IoT-based driver drowsiness systems to enhance the real-world application. S. C et al. [5] applied CNN-facial to analyze fatigue pattern and use IoT cloud connectivity to issue alerts. Nonetheless, a significant number of available systems are characterized by increased latency, reliance on the cloud or impaired multi-level safety responses.

To overcome such constraints, this paper presents a low-latency IoT-based driver fatigue sensor which incorporates the YOLO-based vision processing and an ESP32-controlled multi-level alert system. The suggested solution focuses on


Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on September 25,2026 at 10:06:58 UTC from IEEE Xplore. Restrictions apply.

real-time functionality, the very low lag in communications, and feasibility of commercial automobile safety.

## II. Related Works

The latest studies have also considered expanding the IoT-empowered and multi-modal solutions to support the safety of drivers by generating real-time monitoring and smart alert systems. AlHousrya et al. [6] introduced the use of IoT-driven driver health monitoring device, which incorporates several physiological factors, which include heart rate, eye activity, and breathing with the ESP32 microcontroller. Their cloud portal based model is head-on MQTT and HTTP protocols and analytics based on Azure to detect anomalies. Although the system is shown to be modular and scalable, its use of cloud computation can also be associated with latency, which makes the system unsuitable in cases where fatigue detection needs to occur quickly. Deep learning based on camera has also become a common practice. The CNN-based driver drowsiness detection system suggested by Vijayasudhan and Adarsh [7] is a system in which various visual signals such as eye state, yawning, head tilt, and shoulder posture are considered together. Their system is a system that combines the IoT infrastructure with edge and cloud computing to provide an enhanced detection accuracy and lower false alarms. Nevertheless, real-time responsiveness in embedded systems could be affected by the computational complexity of the multi aspect CNN processing and dependency on the cloud.

To mitigate the challenge of deployment in the various types of vehicles, Naik et al. [8] proposed several drowsiness detection systems, such as IR sensor-based, wearable-based and webcam-based systems. The webcam model is based on traditional vision measurements like Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) with remote alerting through IoT. Despite being a multi-purpose approach, the conventional feature-based approaches can be difficult to use in diverse light and head pose settings than the contemporary deep learning detectors do. In addition to the vision-based solutions, Anand et al. [9] examined EEG-based drowsiness detection that works along with an IoT-based vehicle safety and accident sensor capabilities. Although EEG signals are very reliable in cognitive fatigue detection, they are intrusive, costly and impossible to use on a large scale in a commercial vehicle. Other threats on road safety like drunk driving have been dealt with using IoT as well. As a suggestion, Rahman et al. [10] have suggested an IoT-based drunk driving and accident detection system applying various sensors, GPS, and mobile communication. Though working well at detecting alcohol and response to emergencies, the system fails to respond to visual fatigue indicators important in detecting drowsiness at the first level.

In addition to the vision-based and cloud systems of monitoring, a number of studies have been concentrating on the physiological of sensing and the hybrids of reducing fatigue among drivers and enhancing road safety. Kumar et al. [11] introduced an Anti-Sleep Alarm System (ASAS) that is an IoT-based system based on eye-blink and the monitoring of physiological signals by wearing a headset. The system incorporates automatic braking and alarm systems that will help in avoiding fatigue-related accidents. Although useful to identify fatigue, the wearability of the solution can decrease the level of comfort of drivers, as well as its usability over time, especially in cases of commercial driving. In order to make the safety precautions proactive, S. K et al. [12] presented a hybrid deep learning model that integrates spatial and temporal image information, with in vehicle cameras and sensors to forecast unsafe driving and impending dangers. The system offers both low-latency and better hazard prediction, though as it is now its main objective is driving risk assessment, and not direct fatigue or drowsiness detection, restricting its use to early stage driver alertness monitoring. Combined fatigue and accident detection systems based on integrated IoT and computer vision have also been considered. V. K et al. [13] created the driver monitoring system based on Arduino, which is powered by deep learning, which is used to detect facial drowsiness, and MEMS, which can be used to detect collision. The system will have GPS and GSM modules on emergency alerts. Although it has practical safety response capabilities, the methodology is based on comparatively heavy hardware elements and does not focus on edge communication of low latency. The traditional computer vision methods have been utilized to address lightweight, vision-based fatigue detection. S. K. P. S et al. [14] described a real-time drowsiness detector system on the basis of the eye-blink ratio analysis using OpenCV and dlib. The solution is affordable and non-obstructive but it is sensitive to changes in illumination, occlusions, and changes in head poses and this can lead to poor performance in real world driving environment. K. S. Kumar et al. [15] further explored physiological IoT-based monitoring, proposing a device consisting of IR sensors of eye-blink, pulse sensors, and GPS to identify drowsiness and track the location. Although the system is shown to be effective in real-time alerting and remote monitoring with a web interface, threshold-based detection scheme could not be as robust as learning-based visual inference models.

On the whole, the existing literature indicates the usefulness of physiological, vision-based, and hybrid IoT to driver safety. Nevertheless, the issues surrounding intrusiveness, variability of conditions under which the system can operate, and lower level safety warnings encourage the development of an efficient, non-intrusive, and low-latency IoT architecture with deep learning-based vision processing.

## III. Proposed System

To increase road safety, the driver fatigue detection system proposed combines real-time computer vision, IoT device, and multi-level alert to achieve better control over road safety. Figure 1 shows a proposed work hardware block diagram. The whole architecture is made of three major modules, the vision processing module, the IoT control module and the alerting module.

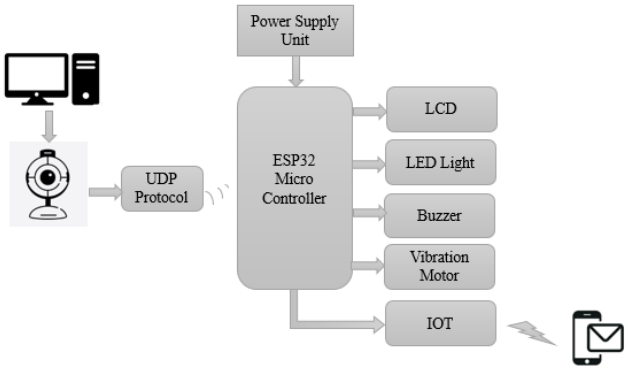



Figure.1 Proposed Work Hardware Block Diagram


Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on September 25,2026 at 10:06:58 UTC from IEEE Xplore. Restrictions apply.

In this system, a high-resolution camera is installed to face the driver in order to record the facial features at all times when using a vehicle. The images obtained are then analyzed with a deep learning model based on YOLOv11 which is trained to identify signs of fatigue such as long eye closing, sudden yawning, and head bobbing.

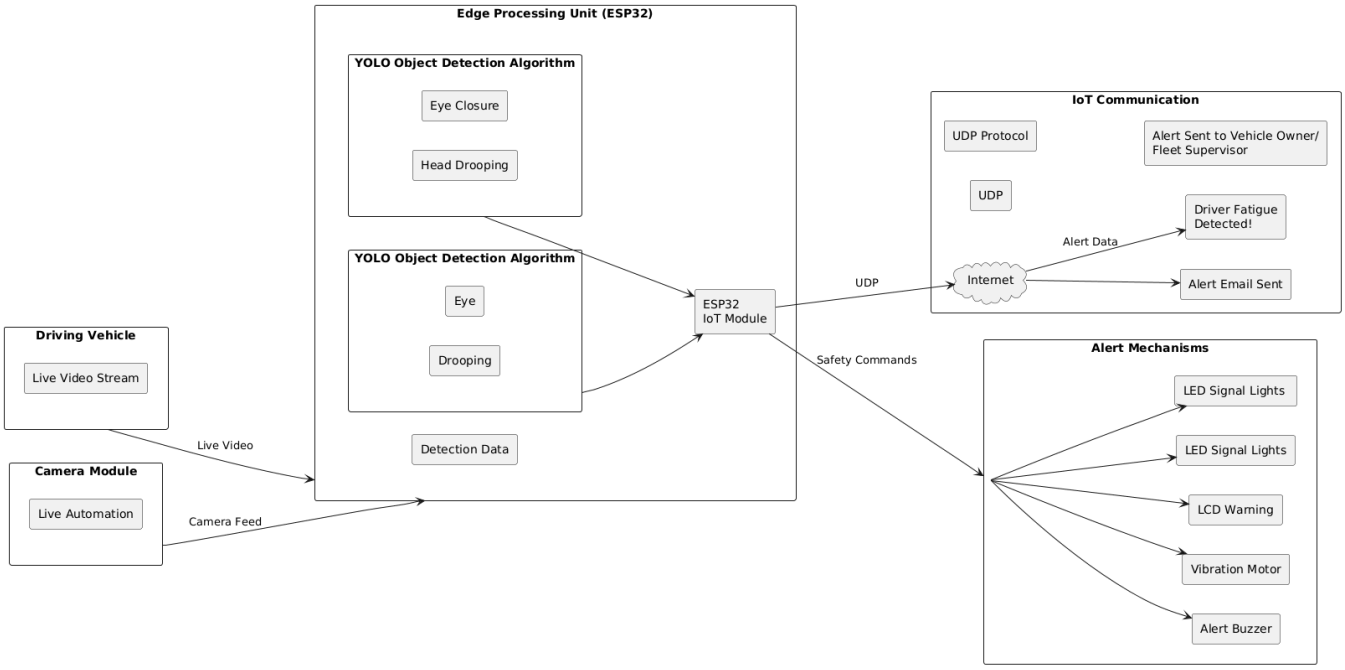



Figure.2 Proposed Work architecture Diagram

YOLO is fast and has high accuracy with detection thus it can be used in dynamic driving applications which require real time. The ESP32 microcontroller is the core component of the IoT, as it interprets the outcomes of the detection provided by the vision module and processes them into relevant safety measures. When the signs of fatigue are detected, the system triggers a multi-tiered alert measure. The initial one has a sound alarm and vibration response to immediately alert the driver. In case the driver fails to react, a second-level alert will be issued via an LCD display showing warning message, with the external LED lights on the car turned on to notify the traffic. Moreover, the system will send an automatic email with the image of the driver to the vehicle owner or fleet supervisor so that they can observe the driver remotely and take appropriate action. The User Datagram Protocol (UDP) is used to guarantee high-speed communication between the processing module and the IoT hardware. UDP reduces the latency due to its connectionless data transfer, which is lightweight in nature and ensures that alert mechanisms are activated almost instantly. Figure.2 shows a proposed work architecture design.

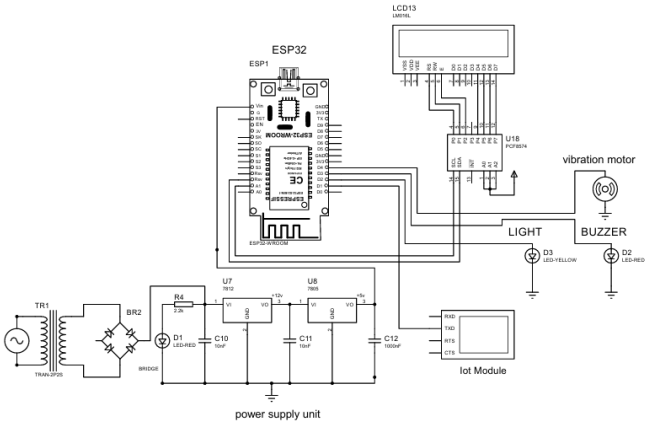



Figure.3 Circuit Diagram

The system design focuses on cost-effective hardware, easy to integrate and to operate in various traffic environments such as night-time driving and long hauls. The suggested framework is an effective and scalable approach to smart transportation systems, commercial fleet management, and personal vehicle safety, which offers real-time detection, instant alerts, and remote monitoring opportunities.

The circuit diagram Figure.3 illustrates the integration of the ESP32 microcontroller with the real-time camera, alert mechanisms, and communication modules for driver fatigue detection. The camera captures the driver's facial region and sends the image data to the processing unit, where the YOLOv11-based model analyzes eye closure and head posture. Upon detecting fatigue, the ESP32 simultaneously triggers multiple safety alerts, including a buzzer, vibration motor, LCD display, and vehicle LED lights. UDP protocol ensures low-latency communication between modules, while Wi-Fi connectivity facilitates automated email alerts to supervisors. The design emphasizes simplicity, cost-effectiveness, and real-time operation for commercial vehicle safety.

## IV. Methodology

The developed methodology will be aimed at approaching the goal of driver fatigue detection in real-time by utilizing computer vision, deep learning, and IoT-enabled alert systems. The system provides low-latency detect and multi-level response systems, which are appropriate in commercial and personal cars. The methodology can be split into three key steps, which are driver monitoring, fatigue detection, and the safety alert activation.


Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on September 25,2026 at 10:06:58 UTC from IEEE Xplore. Restrictions apply.

### A. Driver Monitoring

The first stage involves continuous acquisition of facial images from the driver using a high-resolution camera mounted on the dashboard. The captured video stream is represented mathematically as:

$$\mathrm{F} = \{f_1, f_2, f_3, \ldots, f_n\} \qquad (1)$$

where $f_1$ denotes the $i^{th}$ frame captured at time $t_i$. Each frame undergoes pre-processing to enhance image quality and improve feature detection accuracy. Pre-processing includes grayscale conversion, histogram equalization, and noise reduction. Key facial landmarks, particularly around the eyes, nose, and head region, are extracted for subsequent analysis. The coordinates of these landmarks are represented as:

$$\mathrm{L} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k)\} \qquad (2)$$

where k represents the number of facial landmarks detected per frame.

### B. Fatigue Detection Using YOLOv11

The second stage uses a YOLO-based deep learning model to detect fatigue-related visual cues in real time. Each processed frame $f_i$ is fed into the YOLOv11 network, which outputs a set of detected regions:

$$\mathrm{D}(f_i) = \{(b_1, c_1, s_1), (b_2, c_2, s_2), \ldots, (b_m, c_m, s_m)\} \qquad (3)$$

where $b_j$ is the bounding box for the $j^{th}$ region of interest, $c_j$ is the class label (such as eye closure or head droop), and $s_j$ is the confidence score. The Eye Aspect Ratio (EAR) is calculated to quantify eye closure as follows:

$$EAR = \frac{\left||P_2 - P_6|\right| + ||P_3 - P_5||}{2||P_1 - P_4||} \qquad (4)$$

where $P_1$ to $P_6$ are the eye landmark coordinates. A sustained EAR below a threshold $T_{eye}$ over consecutive frames indicates that the driver is drowsy. Similarly, head posture is analyzed by calculating the tilt angle θ using landmark coordinates:

$$\theta = arctan\frac{y_{nose} - y_{chin}}{x_{nose} - x_{chin}} \qquad (5)$$

A tilt angle beyond a set threshold $T_{head}$ signifies potential fatigue. The combined fatigue score is computed as:

$$s_f = \alpha\left(T_{eye} - \mathrm{EAR}\right) + \beta(\theta - T_{Head}) \qquad (6)$$

where α and β are weight coefficients representing the contribution of eye closure and head tilt, respectively. When $s_f$ exceeds a predefined limit, fatigue is detected.

### C. IoT-Based Safety Alert Activation

Once fatigue is detected, the detection result $D(F_i)$ along with the frame timestamp $t_i$ is transmitted to the ESP32 microcontroller using the User Datagram Protocol (UDP)**:**

$$\mathrm{Packet} = \{\mathrm{D}(f_i), t_i\} \qquad (7)$$

The ESP32 triggers a multi-level alert system. Initially, an audible buzzer and vibration motor are activated to immediately alert the driver. If the driver does not respond, the system escalates the alert through the vehicle's LCD display and external LED lights, notifying surrounding traffic. Additionally, an automated email containing the captured driver image is sent to the vehicle owner or fleet supervisor for remote monitoring.

### D. System Integration and Workflow

The entire system workflow can be mathematically expressed as:

$$\mathrm{Alert} = \mathrm{f}\left(S_f, \mathrm{D}(f_i)\right) \qquad (8)$$

where Alert=1 indicates driver fatigue detection, prompting multi-stage responses. The proposed methodology ensures real-time detection**,** minimal false positives**,** and low-latency alert activation, making it suitable for intelligent transportation systems, fleet management, and personal vehicle safety.

## V. Result & Discussion

### A. System Performance Evaluation

The driver fatigue detection system proposed based on the use of the IoT was tested in varying driving conditions; daytime and night-time conditions, and involved the use of various driver models that portrayed both normal alertness and drowsiness. The most important measures that were evaluated included detection accuracy, latency, and alert response effectiveness. The experiment was carried out using 5,000 video frames that were recorded in real-time driving simulations. The frames were also subjected to running the eye aspect ratio (EAR) and the head droop angle, which were then categorized as either fatigue or non-fatigue events. Table I gives an overview of the detection performance of the YOLOv11 model.

TABLE I. Fatigue Detection Accuracy

| Indicator | True Positives | False Positives | Accuracy (%) |
|---|---|---|---|
| Eye Closure | 1480 | 45 | 97.0 |
| Head Drooping | 1435 | 60 | 95.9 |
| Overall Fatigue | 2915 | 105 | 96.4 |

The findings suggest that the YOLOv11-based detection model is quite efficient to detect fatigue signs. Eye closure and head drooping were also found to be consistent with different lighting conditions and various drivers, and this indicates the stability of the detection mode.

### B. Latency Analysis

Latency was used to determine the period between the capturing of a frame and the triggering of an alert on the ESP32 microcontroller. Three experimental conditions were tested in the form of single alert, multi-alert combination with the use of buzzer, vibration motor, and LED, and email notification. The average latency of each type of alert is showcased in figure 4.


Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on September 25,2026 at 10:06:58 UTC from IEEE Xplore. Restrictions apply.

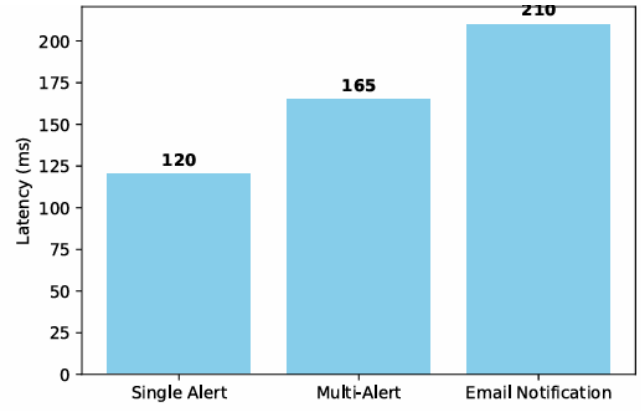



Figure 4. System latency comparison for different alert mechanisms

The average latency of single alerts was 120 milliseconds. Latency was also found to be 165 milliseconds when both multi-level alerts were switched on. A latency of 210 milliseconds was caused by the addition of automated email alerts to the vehicle owner or fleet supervisor. These tests prove the idea that the system can issue warnings in time, and the delay between fatigue monitoring and warning activation is minimal.

### C. *Multi-Level Alert Effectiveness*

The evaluation of the multi-level safety alert system was on the basis of the reaction times of the driver. Table II has shown the mean reaction time of all alert mechanisms.

TABLE II. DRIVER RESPONSE TIME TO MULTI-LEVEL ALERTS

| Alert Type | Average Reaction Time (s) |
|---|---|
| Audible Alarm Only | 3.5 |
| Vibration Motor Only | 3.2 |
| LCD Display Only | 4.0 |
| Multi-Level Alert Combination | 2.1 |

The findings suggest that combined multi-level alert plays a major role in enhancing the responsiveness of the driver. Reaction time was slowest to the combined alert system (down by about 40 % with a single type of alert) indicating the benefit of a hierarchical approach to alert systems in protecting drivers.

### D. *Comparative Analysis with Traditional Systems*

The proposed YOLO-IoT was contrasted with the traditional fatigue detection techniques that are limited to EAR thresholding by not using deep learning. The performance measures were the average detection accuracy and false-positive rate as shown in Figure 5.

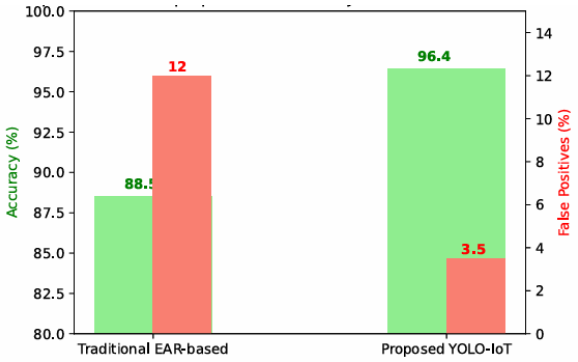



Figure 5 Comparative performance of proposed YOLO-IoT system vs traditional EAR-based system.

EAR-Based systems performed at an accuracy of 88.5% with a rate of false-positive at 12%. On the other hand, the YOLO-IoT system reached an accuracy of 96.4 with a false-positive rate of 3.5%. The given comparison has proven the advantages of combining YOLOv11-based deep learning with IoT-based multi-level alerts that enhance the reliability of the detection much more and minimize the false fatigue alerts.

### E. *Discussion*

The outcomes of the experiment prove that the YOLOv11-IoT driver fatigue detection system can be used to effectively track the driver alertness in real-time. The eye closure and drop of head scores of 96.4 percent indicate high similarity in detection performance in various drivers and lighting situations. System latency (120-210 milliseconds depending on the alert type) is used to be sure that multi-level alerts are triggered in time, i.e. audible, visual, and vibration warnings. Analysis of reaction time represents that multi-level alerts also decrease the driver response time by about 40%, which highlights the importance of these alerts in providing an opportunity to make a corrective action immediately. When compared to the traditional EAR-based systems, it becomes evident that the suggested framework prevents false positives significantly and enhances the overall quality of detection, and there is no need to stress the benefits of the integration of deep learning with IoT-based alert systems. Sensitivity to the light conditions and delays in email notification slightly increased are a limitation. Improvements in the future can be made by using infrared cameras to detect during the night and processing the edges so as to effectively handle large-scale fleets.

## VI. CONCLUSION

The low-latency IoT architecture proposed to implement driver fatigue detection in real time can prove that the combination of computer vision and intelligent alert systems can help to make the road safer. Experimental findings indicate that it has high detection accuracy of 96.4% on the detection of fatigue indicators including long eye closure and head drooping, and has low end-to-end latency of 120-210 ms, depending on the alert level. This fast reaction allows prompting the multi-level safety measures, such as the audible alarms, vibration feedback, LCD notices, and outside vehicle LEDs. The YOLO-based deep learning system results in less false positives and more robust detection in changing driving conditions as compared to more traditional eye-aspect-ratio-based systems. The system is scalable, economical, and perfectly adapted to intelligent transportation and commercial fleet applications besides other support to the remote monitoring by automated email notifications. Future


Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on September 25,2026 at 10:06:58 UTC from IEEE Xplore. Restrictions apply.

directions will be to consider the infrared imaging of better night-time detection, the addition of the edge-based processing to large-scale fleet deployment and the predictive analytics to predict the presence of fatigue. This is likely to further decrease reaction time, improve system reliability in wide range of environment and enable wider application in intelligent vehicle safety systems of the next generation.

## REFERENCES

[1] Y.-S. Poon et al., "Driver Distracted Behavior Detection Technology with YOLO-Based Deep Learning Networks," Proc. IEEE Int. Symp. Product Compliance Eng. – Asia (ISPCE-ASIA), Taipei, Taiwan, pp. 1–5, 2021, doi: 10.1109/ISPCE-ASIA53453.2021.9652435.

[2] M. T. Soe et al., "Abnormal Behavior Detection in Real-time for Advanced Driver Assistance System (ADAS) using YOLO," Proc. IEEE Symp. Industrial Electronics & Applications (ISIEA), Langkawi, Malaysia, pp. 1–6, 2022, doi: 10.1109/ISIEA54517.2022.9873672.

[3] B. Ma et al., "Distracted Driving Behavior and Driver's Emotion Detection Based on Improved YOLOv8 With Attention Mechanism," IEEE Access, vol. 12, pp. 37983–37994, 2024, doi: 10.1109/ACCESS.2024.3374726.

[4] R. Zhao et al., "YOMO-ECA: A Lightweight Multi-Facial Feature Fatigue Driving Detection Method," Proc. 6th Int. Conf. Artificial Intelligence and Computer Applications (ICAICA), Dalian, China, pp. 397–402, 2024, doi: 10.1109/ICAICA63239.2024.10823044.

[5] S. C., L. G., S. A., and Madhan K., "An Advanced Approach for Predicting Driver Drowsiness Using CNN and Real-Time Facial Detection," Proc. Int. Conf. Computational Robotics, Testing and Engineering Evaluation (ICCRTEE), Virudhunagar, India, pp. 1–6, 2025, doi: 10.1109/ICCRTEE64519.2025.11052993.

[6] O. AlHousrya, A. Bennagi, P. A. Cotfas and D. T. Cotfas, "IoT-Enabled Driver Health Monitoring System Using Multi-Modal Sensors and Cloud Integration," Proc. IEEE 31st Int. Symp. Design and Technology in Electronic Packaging (SIITME), Brasov, Romania, pp. 107–114, 2025, doi: 10.1109/SIITME67657.2025.11293652.

[7] G. Vijayasudhan and S. Adarsh, "Enhanced Safety for SDV through Camera-based Driver Drowsiness Detection," Proc. IEEE 6th India Council Int. Subsections Conf. (INDISCON), Rourkela, India, pp. 1–8, 2025, doi: 10.1109/INDISCON66021.2025.11252028.

[8] B. S. Naik, T. V. Khot, B. D. Machado and R. B. Raut, "IoT-Enabled Drowsiness Detection Systems for Enhanced Road Safety Across Diverse Vehicle Types," Proc. Fourth Int. Conf. Power, Control and Computing Technologies (ICPC2T), Raipur, India, pp. 293–298, 2025, doi: 10.1109/ICPC2T63847.2025.10958589.

[9] A. Anand et al., "Real-Time EEG-Based Drowsiness Detection with IoT-Enabled Vehicle Safety Alert System," Proc. 6th Int. Conf. Control, Communication and Computing (ICCC), Thiruvananthapuram, India, pp. 1–7, 2025, doi: 10.1109/ICCC64910.2025.11077179.

[10] A. Rahman et al., "Intelligent Road Safety: IoT-enabled Drunk Driving and Accident Detection System," Proc. IEEE EUROCON 2023 – 20th Int. Conf. Smart Technologies, Torino, Italy, pp. 48–53, 2023, doi: 10.1109/EUROCON56442.2023.10198986.

[11] A. Kumar et al., "Enhanced Driver Safety: IoT Based Anti-Sleep Alarm and Automatic Braking System Using Eye Blink Sensor," Proc. IEEE 21st India Council Int. Conf. (INDICON), Kharagpur, India, pp. 1–6, 2024, doi: 10.1109/INDICON63790.2024.10958220.

[12] S. K, K. A, D. M and D. R. C, "Hybrid Deep Learning for Proactive Driver Risk Prediction and Safety Enhancement," Proc. Int. Conf. Multi-Agent Systems for Collaborative Intelligence (ICMSCI), Erode, India, pp. 1572–1577, 2025, doi: 10.1109/ICMSCI62561.2025.10894603.

[13] V. K et al., "Integrated Driver Monitoring and Real-Time Accident Detection System using IoT and Computer Vision," Proc. 5th Int. Conf. Expert Clouds and Applications (ICOECA), Bengaluru, India, pp. 296–300, 2025, doi: 10.1109/ICOECA66273.2025.00058.

[14] S. K. P. S et al., "Development of Artificial Intelligence System for Driver Drowsiness Detection," Proc. Int. Conf. Multi-Agent Systems for Collaborative Intelligence (ICMSCI), Erode, India, pp. 1812–1816, 2025, doi: 10.1109/ICMSCI62561.2025.10894008.

[15] K. S. Kumar et al., "Real-Time Driver Drowsiness Detection System using IoT-based Physiological Monitoring and Web Interface," Proc. 8th Int. Conf. Computing Methodologies and Communication (ICCMC), Erode, India, pp. 412–416, 2025, doi: 10.1109/ICCMC65190.2025.11140752.


Authorized licensed use limited to: COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY. Downloaded on September 25,2026 at 10:06:58 UTC from IEEE Xplore. Restrictions apply.